ALL YOU NEED TO KNOW ABOUT

MACRONUTRIENTS AND CALORIES

The Power of Nutrition: Using Macronutrients and Caloric Knowledge to Transform Your Life





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Welcome!

Welcome to Enchanted's Weight Loss and Wellness Program! We're excited to be guiding you on your wellness journey. The following information is provided to help you optimize your health by gaining a better understanding of macronutrients and calories.

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What are MACRONUTRIENTS?

Macronutrients are nutrients the body requires in large amounts to maintain proper growth, development, and functioning. The three main macronutrients are carbohydrates, proteins, and fats.

CARBOHYDRATES
Carbohydrates are the body's primary source of energy. They are broken down into glucose, which the body uses for fuel. Carbohydrates can be found in foods like grains, fruits, vegetables, and sugars.

PROTEINS
Proteins are important for building and repairing tissues, making hormones and enzymes, and maintaining a healthy immune system. Proteins can be found in foods like meat, fish, dairy, beans, and nuts.

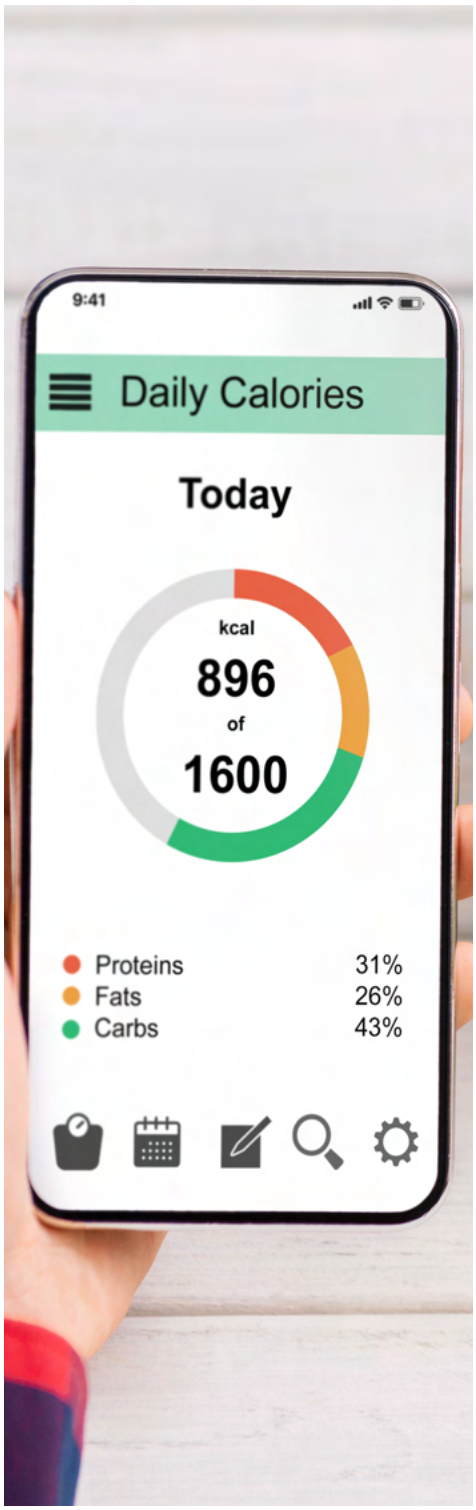
FATS
Fats are important for energy storage, insulation, and cell membrane function. They are also important for the absorption of certain vitamins. Fats can be found in foods like oils, nuts, seeds, and fatty meats.

In addition to these three main macronutrients, fiber is also considered a macronutrient because it is a type of carbohydrate that the body cannot digest. It is important for maintaining proper digestion and bowel function. Fiber can be found in foods like fruits, vegetables, whole grains, and legumes.



What are CALORIES?

Calories are a unit of measurement used to quantify the amount of energy in food. When we eat food, our body breaks it down into its component nutrients, including carbohydrates, protein, and fat, and then uses these nutrients to provide energy for various bodily functions.



Each macronutrient provides a different amount of energy per gram:

- Carbohydrates: 4 calories per gram
- Protein: 4 calories per gram
- Fat: 9 calories per gram

Alcohol, although not a macronutrient, also provides calories at a rate of 7 calories per gram.

The number of calories an individual needs varies depending on age, gender, weight, height, and activity level. Generally, the average adult needs around 2000-2500 calories daily to maintain weight.

It's important to note that while calories are important for weight management, they are not the only factor to consider. Other factors impacting weight include hormonal imbalances, stress levels, sleep quality, and medication use.

Are all the calories CREATED EQUAL?

No, all calories are not created equal. Although a calorie is simply a unit of measurement for energy, the type of food that the calorie comes from can have different effects on the body.

For example, foods that are high in sugar or refined carbohydrates can cause a rapid spike in blood sugar levels, which can lead to a crash and feelings of fatigue and hunger. On the other hand, foods that are high in fiber, protein, or healthy fats can help to slow down the digestion process and provide a more sustained source of energy.

Furthermore, different types of macronutrients require different amounts of energy to be digested and metabolized by the body. For example, protein has a higher thermic effect than carbohydrates or fat, meaning that the body expends more energy to digest and metabolize protein. This means that a calorie from protein may not be the same as a calorie from carbohydrates or fat in terms of the energy required by the body to process it.

In summary, while all calories contain the same amount of energy, the type of food that the calorie comes from and the way it is processed by the body can have different effects on energy levels, metabolism, and overall health.



COUNTING MACROS

vs.

COUNTING CALORIES

You have more control over your appetite

You are more aware of the quality of the food you eat

You can build and maintain muscle and lose body fat more quickly with the proper macro-nutrient ratio

You can adjust your macros to acheive a specific body composition goal

You may experience an increased appetite from imbalanced macros

You risk having nutritional deficiencies if you're not focused on the quality of foods

Inadequate protein will cause you to lose muscle: inadequate carbs will affect your performance; not enough fat will negatively affect hormones

You can adjust your calories to achieve a specific body weight





More about
PROTEINS

Proteins have a caloric value of 4 calories/gram and can be of plant or animal origin. They are vital for growth, muscle tissue repair, immunity, hormone and enzyme production, and muscle mass.

The best part? Its nutritional value doesn't end there! Other benefits of this building-block compound include:

- Healthy skin, hair, and nails
- Strong bones
- Lasting full-feeling
- Boosts immune system

Protein is found in meats, cheeses, milk, nuts, and even vegetables or grains/seeds. The percentage of protein in each food varies widely, from 15% for cheese to 50% for lamb or 40% for beef.

40g protein x 4 calories/gram = 160 calories



More about
CARBOHYDRATES

Carbohydrates have a caloric value of 4 calories/gram and have mainly plant origin. Carbohydrates provide the vital energy that keeps us moving daily, supporting the functions of the nervous, kidneys, and muscular systems, including the heart.

Carbohydrates make a vital contribution to the digestive system because they have a large amount of fiber. The body does not digest fibers, and with their elimination, they help to clean the digestive system. Carbohydrates are found in grains (bread, pasta, rice), legumes, vegetables, and fruits.

Carbohydrates comprise three components: fiber, starch, and sugar. Fiber and starch are complex carbs, while sugar is a simple carb. Depending on how much of each of these is found in food determines its nutrient quality.

The percentage of carbohydrates also varies by food. For example, a banana has about 25g of carbohydrates, while a large potato has 50g.

25g carbs x 4 calories/gram = 100 calories



More about FATS

Fats have the highest caloric value per gram, 9 calories/gram, and even though they have a bad reputation, they are vital to our bodies. Fats help the body grow and develop, absorb vitamins, maintain tissue membranes, provide energy, and aid digestion, which is vital for liver functions.

Fats are found in some types of meat, nuts, cheeses, butter, oils, and fish.

It should be noted that there are three types of fat: **saturated** (found in meat, animal fat, and butter), **trans** (found in processed food, fast foods, and chips), and **unsaturated** (found in olive oil, avocados, and nuts). The first two, saturated and trans, are shown to contribute to heart disease.

To give you an idea, an avocado contains 25g of unsaturated fat, which is very good for you and covers 40% of the daily fiber requirement.

25g carbs x 9 calories/gram = 225 calories



Macronutrients CHEATSHEET for Proteins

FOOD	CAL	PROTEINS	CARBS	FATS
Beef, ground (80% lean)	254	17g	0g	20g
Beef, filet mignon	271	20g	0g	8g
Beef, sirloin steak	250	26g	0g	10g
Chicken breast, skinless	165	31g	0g	1g
Chicken thighs, skinless	209	26g	0g	9g
Turkey breast, skinless	104	29g	0g	1g
Turkey, ground (80% lean)	171	18g	0g	10g
Pork chop, center loin	221	26g	0g	9g
Pork tenderloin	143	23g	0g	3g
Pork, ground (80% lean)	287	18g	0g	14g
Salmon, wild-caught	206	20g	0g	13g
Sardines	208	25g	0g	11g
Tilapia	96	26g	0g	2g
Tuna	116	30g	0g	1g
Shrimp	85	24g	0g	1g
Butter	717	0.9g	0g	81g
Cheddar cheese	403	25g	1.2g	33g
Colby cheese	394	23g	1.6g	29g
Cottage cheese (regular)	342	6g	4g	34g
Cottage cheese (low-fat)	72	10g	2.8g	1g
Cream cheese (reguar)	342	3g	3g	34g
Cream cheese (low-fat)	246	7g	3g	10g
Feta cheese	264	14g	1g	21g
Gouda cheese	356	25g	2g	28g
Yogurt, Greek (regular)	97	10g	4g	10g
Yogurt, Greek (low-fat)	73	9g	3.6g	2g
Eggs	155	13g	1.1g	11g
Tofu	76	8g	2g	6g
Tempeh	193	19g	9g	11g

*All values are per 100g of the product

Macronutrients CHEATSHEET for Carbohydrates

FOOD	CAL	PROTEINS	CARBS	FATS
Oats	389	11g	66g	6.5g
Brown rice	111	2.7g	23g	2.6g
Quinoa	120	4.4g	21g	6g
Barley	354	2.3g	28g	1.2g
Whole wheat bread	247	9.4g	49g	2.4g
Whole wheat pasta	131	13g	26g	1g
Buckwheat	343	13.3g	71g	2.7g
Rye	335	6g	79g	1.5g
Sweet potatoes	86	1.6g	20g	0.1g
Potatoes	87	2g	17g	0g
Corn	96	3.2g	19g	1.1g
Peas	81	5.4g	14g	0.4g
Pumpkin	26	1g	7g	0.1g
Squash	45	1g	4g	0.1g
Yams	118	1.5g	28g	0g
Lentils	116	9g	20g	1g
Chickpeas	364	8.9g	27g	6g
Black beans	341	8.9g	23g	0.9g
Kidney beans	127	8.7g	22g	1.1g
Navy beans	347	8.2g	24g	0.1g
Lima beans	338	7.8g	15g	0.3g
Apples	52	0.3g	14g	0.1g
Bananas	89	1.1g	23g	0.3g
Oranges	47	0.9g	12g	0.2g
Grapes	69	0.7g	18g	0.1g
Melons	34	0.8g	8g	0.1g
Peaches	37	0.9g	10g	0.2g
Pears	57	0.4g	15g	0.1g
Mangoes	60	0.8	15g	0.3g

*All values are per 100g of the product

Macronutrients CHEATSHEET for Fats

FOOD	CAL	PROTEINS	CARBS	FATS
Butter	717	0.8g	0.6g	81g
Ghee	900	0g	0g	99g
Margarine	717	02.g	0.3g	80g
Mayonnaise	684	1.1g	3.2g	75g
Olive oil	884	0g	0g	100g
Coconut oil	862	0g	0g	100g
Avocado oil	884	0g	0g	100g
Flaxseed oil	884	0g	0g	100g
Avocado	160	2g	8.5g	15g
Almonds	579	21.2g	21.6g	49g
Walnuts	654	15.2g	13.7g	65g
Pecans	691	9g	3.9g	72g
Cashews	553	18.2g	30g	44g
Macadamia nuts	718	7.9g	4.5g	76g
Brazil nuts	656	14.3g	2.3g	66g
Peanut butter	589	25g	20.6g	50g
Almond butter	614	21.2g	18.7g	50g
Flaxseeds	534	18.2g	28.8g	42g
Chia seeds	486	16.5g	42g	31g
Sunflower seeds	584	20.7g	20.7g	51g
Pumpkin seeds	559	30.2g	10.7g	49g
Hemp seeds	553	31.5g	3.4g	49g
Tahini	633	17g	17g	53g
Olives	115	0.8g	3.8g	15g
Dark chocolate	604	5.5g	45g	30g

*All values are per 100g of the product

COOKED *vs.* UNCOOKED





Boiled food (rice, pasta, oats and some veggie) tend to absorb water and ↑weight and size





Protein sources (meat) tend to lose water and ↓weight and size

- Weight changes are due to water loss or water absorption. When the weight changes, the calories, and nutrition do not change. 6oz/170g of cooked chicken contains the same calories as 8oz/220g of raw chicken; it simply loses water weight.

- Nutrition labels **will always list the raw weight** unless expressly stated otherwise.

- If you are using a food tracking app, you can search for a "**cooked**" version to avoid conversions for certain foods.

- Weighing raw food will always yield the most consistent & accurate results, but since that is not always possible, you can effectively weigh and track cooked food.

Calculate Your MACRONUTRIENTS

Calculating your macronutrient recommendations involves determining your daily calorie needs and then allocating a percentage of those calories to each macronutrient: carbohydrates, protein, and fat.

Here are the general steps to calculate your macronutrient recommendations:

- Calculate your daily calorie needs: This can be done using an online calculator or through a formula that takes into account your age, gender, weight, height, and activity level.

- Determine your protein needs: Protein needs vary depending on factors such as age, gender, and activity level. Aim for 0.8-1 gram of protein per pound of body weight as a general guideline.

- Allocate calories to carbohydrates: Carbohydrates provide energy and should make up the bulk of your calorie intake. A good starting point is to aim for 45-65% of your total daily calories to come from carbohydrates.

- Allocate calories to fat: Fat is important for hormone production and other bodily functions. Aim for 20-35% of your total daily calories to come from fat.

Fill in any remaining calories with your macronutrient of choice: After determining your protein, carbohydrate, and fat needs, you may have some remaining calories to allocate to a macronutrient of your choice, such as fiber or alcohol.

EXAMPLE ON THE NEXT PAGE

Let's say that you are a 35-year-old male who weighs 170 pounds and is 5'11" tall. You work out three to four times weekly and have a moderate activity level.

To calculate your macronutrient recommendations, you can follow these steps:

1. Calculate your daily caloric needs using a BMR calculator (more details on how to calculate it on page 17). Let's say your BMR is 1,800 calories.

2. Factor in your activity level. If you have a moderate activity level, you can multiply your BMR by 1.55. This would give you a daily caloric requirement of 2,790 calories (1,800 x 1.55 = 2,790).

3. Determine your desired macronutrient ratios based on your goals. Let's say you want to follow a balanced diet with a macronutrient ratio of 40% carbohydrates, 30% protein, and 30% fat.

4. Calculate your daily macronutrient intake. To do this, multiply your total daily caloric intake (2,790) by the percentage of each macronutrient you want to consume. In this example, you would need:

- 1,116 calories from carbohydrates (40% of 2,790)
- 837 calories from protein (30% of 2,790)
- 837 calories from fat (30% of 2,790)

Activity Level Factors

1. Sedentary (little or no exercise): BMR x 1.2
2. Lightly active (light exercise): BMR x 1.375
3. Moderately active (moderate exercise): BMR x 1.55
4. Very active (hard exercise or sports 6-7 days a week): BMR x 1.725
5. Super active (very hard exercise or sports, physical job): BMR x 1.9

Calculate Your BMR

BMR stands for Basal Metabolic Rate, which is the amount of energy (in calories) that your body needs to maintain essential functions while at rest. Calculating your BMR can help you determine how many calories your body needs to function correctly.

There are several formulas to calculate BMR, but one of the most commonly used formulas is the Harris-Benedict equation. The formula differs for men and women, considering factors such as height, weight, age, and sex.

FOR MEN:
BMR = 88.362 + (13.397 x weight in kg) + (4.799 x height in cm) - (5.677 x age in years)
FOR WOMEN:
BMR = 447.593 + (9.247 x weight in kg) + (3.098 x height in cm) - (4.330 x age in years)

To use the formula, you'll need to know your weight in kilograms, height in centimeters, and age in years. Once you've calculated your BMR, you can use it to determine how many calories you must consume daily to maintain, gain, or lose weight.





Thank you

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